# DIFFERENTIALS OF A SMOOTH MORPHISM OF ALGEBRAIC SPACES ARE FINITE LOCALLY FREE

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 1279-1285 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 1287-1292, the complete author proof (4 body lines).

*Editorial note.* Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* The schemes statement itself lives in the Morphisms chapter, outside both of this lane's pools, so no double counting arises.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The etale-locality of statements about the module of differentials of a morphism of spaces (same chapter, lemma-localize-differentials) and the schemes case (Morphisms, lemma-smooth-omega-finite-locally-free); cited inputs with the reduction written in the retained span.

*Source reference.* Original lemma 04CU: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 981–996. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 02G1: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1292–1292. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $S$ be a scheme. Let $f : X \to Y$ be a smooth morphism of algebraic spaces over $S$. Then the module of differentials $\Omega_{X/Y}$ is finite locally free.*

*Proof.* The statement is étale local on $X$ and $Y$ by Lemma 04CU. Hence this follows from the case of schemes, see Morphisms, Lemma 02G1. □

*Date*: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.